

Problem Statement

Bangla, despite being spoken by over 265 million people worldwide, remains a low-resource language in NLP research. The July Revolution of 2024 in Bangladesh generated massive public discourse across social media platforms, reflecting complex public sentiment that is valuable for researchers and policymakers to understand. However, building effective sentiment analysis systems for Bangla is challenging due to its complex morphology, code-switching tendencies, and scarcity of annotated datasets. Most existing work either focuses on non-political domains or uses single transformer models without exploring hybrid architectures. This project addresses that gap by systematically evaluating and comparing multiple machine learning approaches on the Student-People Mass Uprising Public Sentiments Dataset, a 4,200-comment Bangla social media corpus related to the July Revolution, to identify the most effective sentiment classification strategy for this domain.

Project Objectives

The key objectives of this project are:

1. Perform exploratory data analysis to understand the dataset's structure, class distribution, platform-wise sentiment patterns, and data quality issues.
2. Preprocess the Bangla text by handling duplicates, removing noise, and normalizing text for model training.
3. Extract features using TF-IDF and transformer-based embeddings from BanglaBERT.
4. Implement and evaluate at least five machine learning classifiers including Logistic Regression, SVM, Random Forest, XGBoost, and an ensemble method.
5. Compare model performance using accuracy, precision, recall, and F1-score to identify the best-performing configuration.
6. Interpret results and document findings in a reproducible, well-organized codebase and project report.

Literature Review

Bangla, spoken by over 265 million people worldwide, remains critically underrepresented in Natural Language Processing research despite being one of the most widely spoken languages in the world. Sen et al. (2022) provide a foundational overview of this challenge through a comprehensive survey of Bangla NLP covering 75 research papers across eleven domains including sentiment analysis, machine translation, and speech recognition. Tracing the field's evolution from the late 1980s, the authors systematically evaluate classical methods such as SVM, Naive Bayes, and Decision Tree alongside deep learning architectures including CNN, RNN, LSTM, and GRU. Their central finding is that while deep learning approaches show considerable promise, the field remains severely constrained by the complexity of Bangla grammar and a persistent scarcity of publicly available annotated resources. This survey effectively establishes the broader landscape within which all subsequent work in Bangla sentiment analysis, including our own, must be situated.

Within this resource-scarce environment, early work on Bangla sentiment analysis relied heavily on character-level and word-level neural approaches. Haydar et al. (2017) addressed this by proposing a character-level supervised Recurrent Neural Network using GRU units for Bangla sentiment classification, arguing that character-level modeling is better suited to the morphological complexity of Bangla than word-level approaches. Using a dataset of over 45,000 Facebook comments from e-commerce and telecom sectors, labeled as positive, negative, and neutral, their system achieved approximately 84% accuracy, outperforming word-level RNN baselines at 77%. Importantly, the system demonstrated robustness on noisy social media text without requiring extensive NLP preprocessing pipelines.

While this work predates the transformer era, it establishes the foundational insight that Bangla social media text requires language-aware modeling strategies, a principle that continues to guide more recent transformer-based research.

The introduction of pretrained transformer models marked a turning point for Bangla NLP. Bhattacharjee et al. (2022) introduced BanglaBERT, a BERT-based language model pretrained on a 27.5 GB corpus called Bangla2B+, compiled by crawling 110 popular Bangla websites and containing over 2.18 billion tokens. Unlike standard masked language modeling, BanglaBERT was pretrained using the ELECTRA objective, which trains a generator and discriminator jointly, offering substantially better computational efficiency. Benchmarked across four downstream NLU tasks covering sentiment classification, natural language inference, named entity recognition, and question answering under the first-ever Bangla Language Understanding Benchmark (BLUB), BanglaBERT achieved a score of 77.09, outperforming multilingual models like mBERT and XLM-R base by significant margins. Critically for our project, BanglaBERT also demonstrated superior sample efficiency with fewer than 1,000 training examples, making it a well-justified choice as a feature extractor for our 4,200-sample dataset.

Building on BanglaBERT's strong representations, subsequent research explored how transformer models can be extended and combined to push Bangla sentiment classification further. Islam and Alam (2024) constructed BangDSA, a large-scale dataset of 203,463 comments collected from Facebook, YouTube, TikTok, Instagram, and Likee, and evaluated 21 hybrid feature extraction techniques. Their proposed skipBangla-BERT method, combining BanglaBERT embeddings with Skipgram, outperformed all other approaches, while a hybrid CNN-BiLSTM model achieved 95.71% accuracy on three-category classification. Similarly, Mahmud et al. (2025) proposed

a pipeline that integrates a novel rule-based Bangla Sentiment Polarity Score algorithm with BanglaBERT, achieving 89% accuracy on Bengali e-commerce reviews from Daraz Bangladesh, surpassing the standalone BanglaBERT pipeline at 79%. Together, these works confirm that augmenting BanglaBERT with complementary representations, whether statistical, lexical, or neural, consistently yields stronger results than using the model in isolation.

This same principle of combining multiple transformer models rather than relying on a single one is further validated by Hoque et al. (2024), who evaluated five transformer models including mBERT, BanglaBERT, Bangla-BERT-Base, DistilmBERT, and XLM-RoBERTa-base on a Bengali social media dataset of 11,807 comments. After systematic hyperparameter tuning, they proposed a Transformer-ensemble model that achieved 95.97% accuracy with an F1-score of 95.96%, substantially outperforming each standalone model. This finding directly supports the hybrid architecture adopted in the paper that forms the methodological foundation of our project. Hossen et al. (2025) built on precisely this logic by proposing XMB-BERT, a hybrid model that concatenates embeddings from mBERT, BanglaBERT, and XLM-RoBERTa, and evaluated it on a manually curated dataset of 4,200 balanced Bangla social media comments collected during and after the July Revolution in Bangladesh. Paired with a voting classifier and PCA-based dimensionality reduction, XMB-BERT achieved 83.7% accuracy across positive, negative, and neutral classes, outperforming all individual transformer baselines. This dataset forms the foundation of our current project.

The importance of domain-adaptive pretraining as an extension of the core BanglaBERT model is demonstrated by Joy et al. (2023), who introduced BanglaClickBERT for Bangla clickbait detection in news headlines. By performing domain-adaptive pretraining on one million

unlabeled headlines before fine-tuning, and concatenating the resulting representations with XLM-RoBERTa through a Multi-Layer Perceptron, the authors achieved a state-of-the-art F1-score of 75.51%, outperforming both statistical and deep learning baselines. While clickbait detection differs from sentiment analysis, this work contributes a methodologically relevant insight: that adapting BanglaBERT to specific domains and combining it with cross-lingual models significantly improves classification performance for low-resource Bangla text, a pattern directly aligned with the hybrid strategy employed in our approach.

Finally, extending the scope of Bangla social media text classification beyond sentiment, Hasan et al. (2025) introduced BanglaMultiHate, one of the largest manually annotated Bangla datasets comprising approximately 51,000 YouTube comments labeled across three simultaneous tasks: type of hate, severity, and target. Their comprehensive benchmarking of SVM, BanglaBERT, and large language models including Llama3 and Qwen3 under zero-shot and LoRA fine-tuning settings consistently found that fine-tuned BanglaBERT outperforms all other models, while zero-shot LLMs fail to exceed even the majority baseline. This result reinforces a recurring conclusion across the literature: for socially and politically charged Bangla text on social media, language-specific pretraining through BanglaBERT remains the most reliable approach, and transformer-based classifiers built on such representations generalize well to diverse classification tasks beyond binary sentiment, including the three-class classification task central to our project.

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Dataset

Dataset Name: Student-People Mass Uprising Public Sentiments Dataset

Source: <https://doi.org/10.5281/zenodo.15342899>

Format: CSV (.csv), 4 columns (comment, language, platform, label)

Description: This dataset contains 4,200 manually collected Bangla-language social media comments gathered between September and October 2024 from Facebook, YouTube, and Twitter (X) during and after the July Revolution in Bangladesh. Comments are equally distributed across three sentiment classes :Positive, Negative, and Neutral with 1,400 samples per class, making it a perfectly balanced dataset. Each comment is labeled based on its tone: supportive comments toward the protest movement are marked Positive, comments containing slang or threats directed at protesters are marked Negative, and comments with neither support nor offense are marked Neutral. The dataset was introduced by Hossen et al. (2025) alongside their hybrid XMB-BERT sentiment analysis framework and is publicly available for research purposes.